

Multi-scene setup: U-ESfM vs RESfM

How each method trains across scenes and evaluates per scene, and where the configurations differ

How RESfM's multi-scene training is set up (paper + released code)

RESfM's goal is a single model that generalizes across scenes, with outlier robustness supplied by a supervised classifier. The training data are 27 MegaDepth scenes (each under 1000 images), with 4 further scenes held out for validation. Each scene is represented by a point-track matrix built by concatenating pairwise SIFT matches filtered with RANSAC; these tracks realistically contain roughly 12-44 percent outliers, and - crucially - ground-truth outlier labels are available for the training scenes. In every epoch, training visits all 27 scenes (the released code groups four scenes per optimizer step), and from each scene it samples a random subset of 10-20 percent of the cameras, so the network constantly sees small, varied sub-problems. The architecture is the shallow set-of-sets equivariant network: a shared 3-layer encoder with 256 feature channels, plus three heads - cameras, 3D points, and an outlier classifier. The loss combines two terms: a reprojection loss in which each correspondence is down-weighted by the classifier's predicted outlier probability, and a class-balanced binary cross-entropy that trains the classifier against the ground-truth labels (weighted 0.1 relative to the reprojection term). The supervised term is what anchors the classifier - without it, the weighting term alone would reward calling everything an outlier. Optimization uses Adam with normalized gradients at learning rate $1e-3$, a budget of 20,000 epochs with mild learning-rate decay (factor 0.9 at milestones from 10,000 onward), evaluation on the four full validation scenes every 500 epochs, and best-checkpoint selection by outlier-classification accuracy on validation. The seed is fixed to 20; the paper reports roughly 16 hours on a single A40 GPU, and the released model is the best checkpoint, from epoch 18,500.

How RESfM evaluates on a test scene

For each of the 36 test scenes, the trained model runs feed-forward on the complete point tracks; its classifier's outlier scores are thresholded at 0.6, flagged observations are removed, and if the removals disconnect the scene's view graph only the largest connected component is kept. The camera and point heads are then fine-tuned on the pruned tracks for 1,000 epochs with the plain reprojection loss (the classifier is frozen), and the result goes through the paper's Robust Bundle Adjustment: a first BA pass, removal of 3D points with reprojection error above 5 pixels or seen by fewer than 3 cameras, largest-component extraction, re-triangulation, and a second, standard BA. Table 1's per-scene numbers (registered cameras, rotation and translation errors) come out of this pipeline.

How U-ESfM's multi-scene training is set up

U-ESfM keeps RESfM's entire protocol scaffolding and removes the supervision. Stage-1 training uses exactly the same data and sampling: the same 27 training and 4 validation MegaDepth scenes, the same outlier-contaminated tracks, the same per-epoch visiting of all scenes with 10-20 percent camera subsets, the same grouping of four scenes per optimizer step, and the same seed (20). Two things change by design. First, the architecture is the project's Deep equivariant network - two blocks of three layers each (256 channels) with residual connections, sparse layer normalization, and sparse dropout - the report's finding being that depth, properly regularized, is the primary performance driver. Second, the loss is the plain unsupervised reprojection loss alone, computed only from the predicted cameras and points against the observed tracks; no outlier classifier participates in training and no labels are touched. Optimization uses Adam with normalized gradients at learning rate $1e-4$ (chosen by a head-to-head comparison after RESfM's $1e-3$, tuned for their shallow model, plateaued on the deep one), the same 20,000-epoch budget, the same every-500-epochs validation cadence, and best-checkpoint selection by validation reprojection error - an unsupervised criterion, where RESfM's accuracy criterion requires labels. Training runs in an environment matched to RESfM's release (same PyTorch and pycolmap versions), with checkpointing wired so a preempted run resumes losslessly.

How U-ESfM evaluates on a test scene

The evaluation mirrors RESfM's stage for stage, with the learned classifier replaced by robust statistics. The trained stage-1 model runs feed-forward on the scene's complete tracks and its per-observation reprojection errors are collected; outliers are flagged by the MAD rule - error above the scene's median plus 2 times 1.4826 times the median absolute deviation - which adapts automatically to each scene's error scale and requires no labels ($\delta = 2$ is the report's best-performing setting). The flagged observations then enter the identical pruning path (threshold gate, largest connected component), the identical 1,000-epoch fine-tune of cameras and points with the reprojection loss, and the identical Robust Bundle Adjustment with the paper's 5-pixel filter. This path was validated end-to-end with RESfM's own released checkpoint before being adopted, where on the smoke-test scene it matched the paper's published numbers. The final report uses seed 20 (the papers' protocol seed) plus a median over five seeds.

Differences in training configuration, summarized

Identical by construction: the scene split (27 train, 4 validation), the point tracks and their outlier content, per-epoch scene visiting with 10-20 percent camera subsampling, four scenes per optimizer step, Adam with normalized gradients, the 20,000-epoch budget, evaluation every 500 epochs with best-checkpoint retention, the fixed seed 20, the 1,000-epoch per-scene fine-tune, and the Robust BA with the 5-pixel filter. Different by design: (1) supervision - RESfM trains its outlier classifier with ground-truth labels; U-ESfM uses no labels anywhere; (2) the loss - classifier-weighted reprojection plus balanced BCE versus plain reprojection; (3) the architecture - shallow 1x3 versus Deep 2x3 with residuals, sparse layer normalization, and dropout; (4) test-time outlier identification - learned classifier scores thresholded at 0.6 versus per-scene MAD statistics on reprojection errors; (5) the model-selection metric - label-based classification accuracy versus unsupervised validation reprojection error. Small deviations worth recording: the learning rate (1e-4 versus RESfM's 1e-3, justified empirically for the deep model), the hinge-loss margin in the reprojection loss (1e-4, inherited from ESfM and the project's single-scene practice, versus 1e-3 in RESfM's released configuration), and slightly different learning-rate milestone lists at the same decay factor (negligible at 0.9). Wall-clock differs for mechanical reasons: RESfM reports about 16 hours; U-ESfM's deep model plus full determinism settings run several times slower per epoch.

A note on the shelved learned-classifier variant

An earlier U-RESfM variant kept RESfM's classifier head and replaced the supervised BCE with self-generated pseudo-labels (confident inliers and outliers taken from the tails of the current reprojection-error distribution). At multi-scene scale this collapsed: because the weighted reprojection term rewards declaring every observation an outlier (which zeroes that loss), and the pseudo-label term was too weak to prevent it, the classifier saturated at predicting outlier probability 1 everywhere - which also silently removed the training gradient from the camera heads. The diagnosis is preserved with the run records; a candidate structural fix (detaching the classifier's scores inside the weighting term) exists but is deliberately parked, since the statistical route both matches the project report's primary method and avoids the failure mode entirely.